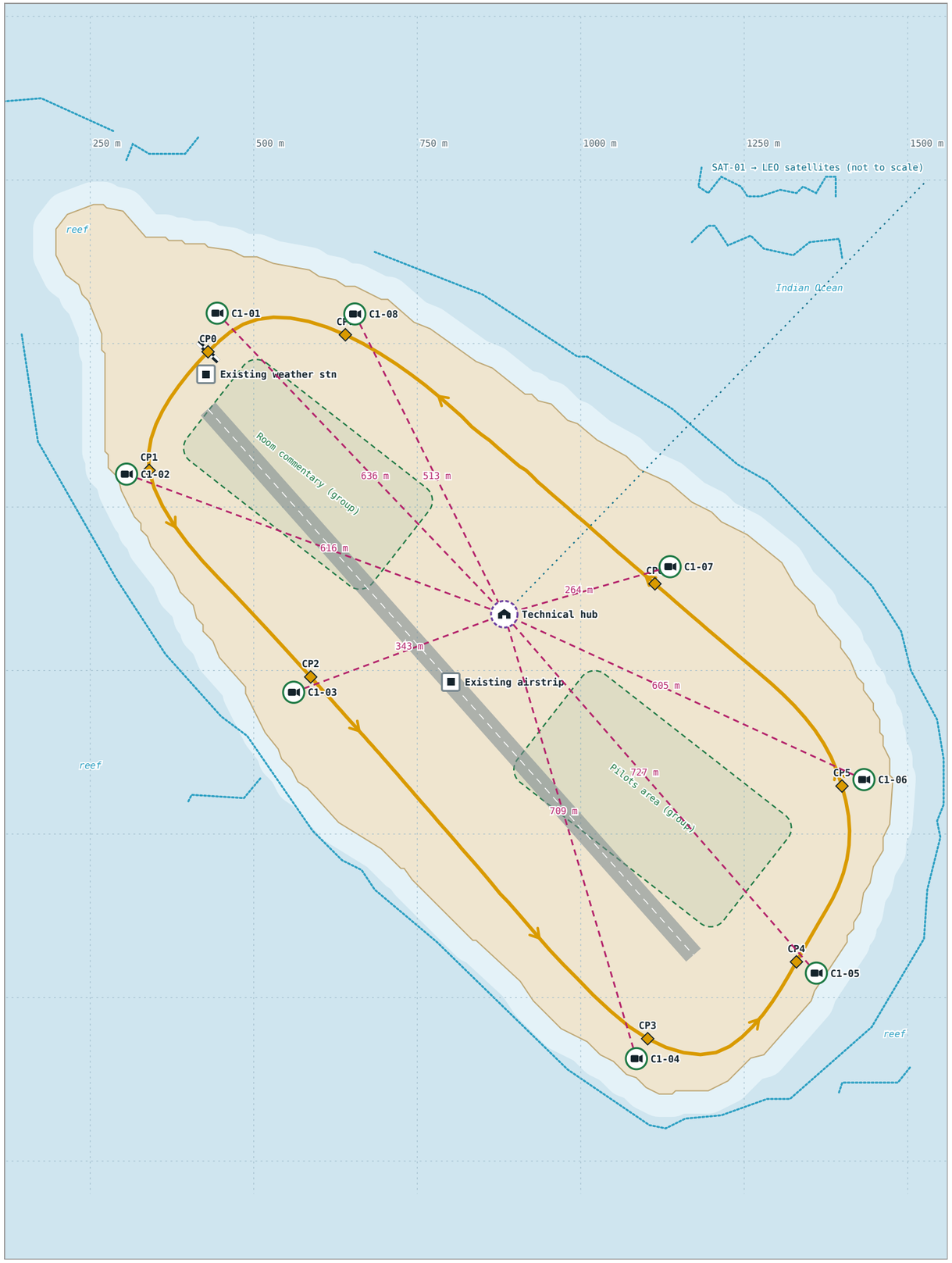
Tromelin · radio solution for the cameras · LOW-COST: IP radio

v0.4 · 30/10/2026 · Preliminary plan, not a topographic survey · Base: CIA World Factbook map, 1 px = 5 m · Symbols enlarged · Straight-line distances, ±15%



How it works

- Each camera C1-01…C1-08 sends its video **by radio** (dashed lines) to mast MST-01 at the technical hub.
- Each camera has an IP radio (RL-xx); a base station (RL-BS) on the mast receives all 8.
- In the control room the mixer combines cameras, 4 drones and commentary; graphics add positions, heart rate and weather.
- A single Starlink (SAT-01) takes the live programme off the island: it is the mandatory exit to the audience. Iridium (SAT-BK) carries backup data only.

Radio links

Camera	Films	Distance	Estimated signal	Antenna at least
C1-01	CP0 start/finish	636 m	-47.3 dBm	2.8 m
C1-02	CP1	616 m	-47.0 dBm	2.7 m
C1-03	CP2	343 m	-42.0 dBm	2.3 m
C1-04	CP3	709 m	-48.3 dBm	2.9 m
C1-05	CP4	727 m	-48.5 dBm	2.9 m
C1-06	CP5	605 m	-46.9 dBm	2.7 m
C1-07	CP6	264 m	-39.7 dBm	2.1 m
C1-08	CP7	513 m	-45.4 dBm	2.6 m

Calculated at 5.5 GHz with 1 W radiated and a 26 dBi receiving antenna. A signal above about −65 dBm is considered strong. “Antenna at least” = height above shrubs and people to keep the line of sight clear.

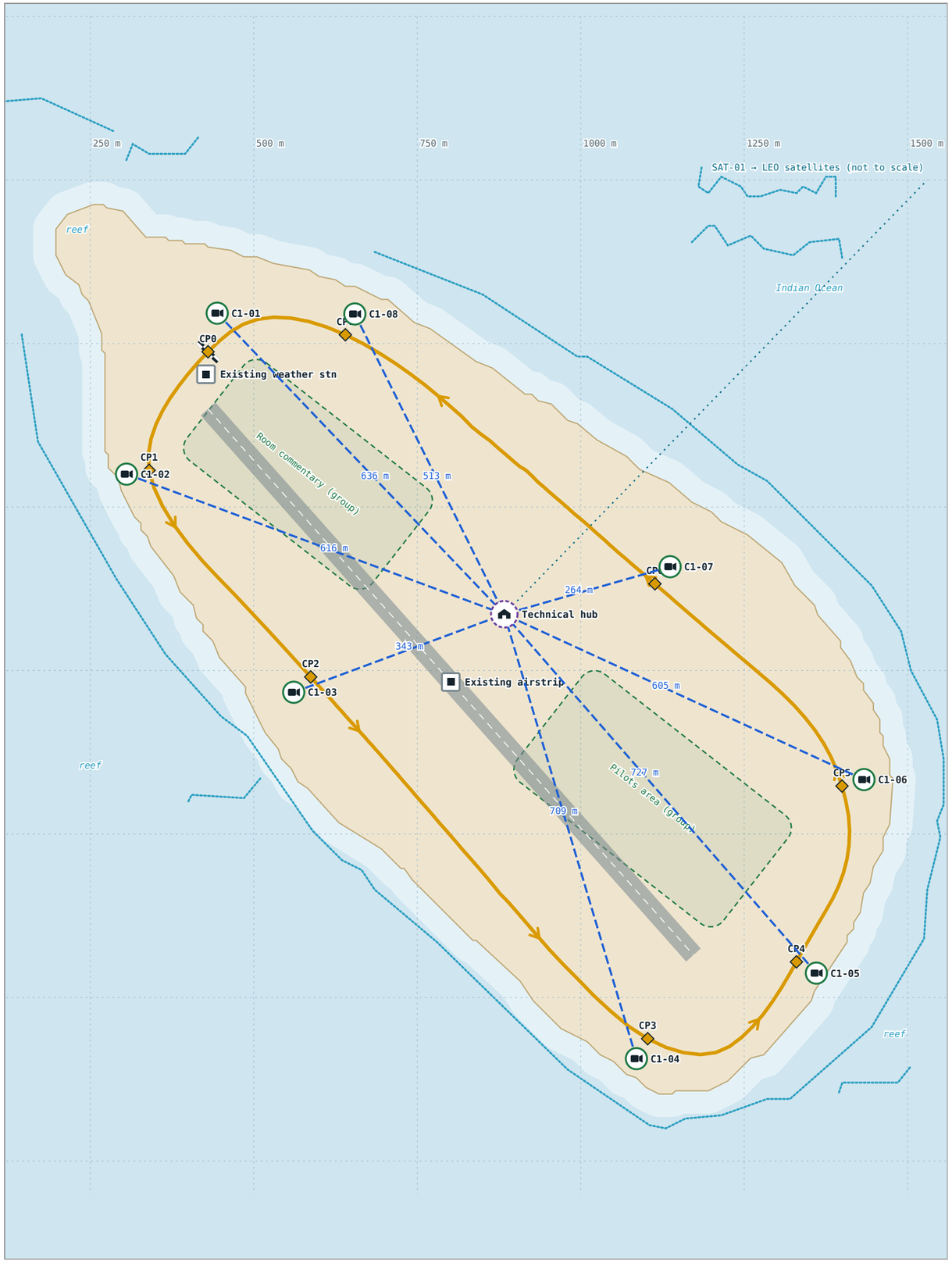
Key data

- Track lap: ~3.23 km · technical hub 130 m from the airstrip and 187 m from the track.
- Flat island: maximum ~6 m, shrubs 0.5–1 m.
- Heavy rain: −0.3 dB on the longest link, negligible.
- Capacity: ~600 Mbps from the base, ~80 Mbps needed.
- Delay: a few milliseconds.
- Radio network cost: base $441 + 8 radios at AUD 231, plus antenna and masts.

Legend: yellow = race track · dashed magenta = IP radio · dotted light blue = towards the satellites (not to scale) · diamond = checkpoint · green = group position · dashed purple = proposal · grey = existing, not verified.

Tromelin · radio solution for the cameras · PREMIUM: Vislink video transmitters

v0.4 · 30/10/2026 · Preliminary plan, not a topographic survey · Base: CIA World Factbook map, 1 px = 5 m · Symbols enlarged · Straight-line distances, ±15%



How it works

- Each camera C1-01…C1-08 sends its video **by radio** (dashed lines) to mast MST-01 at the technical hub.
- Each camera has a Vislink HCAM video transmitter (VTX-xx); 4 Quantum receivers (VRX) on the mast, 2 streams each.
- In the control room the mixer combines cameras, 4 drones and commentary; graphics add positions, heart rate and weather.
- A single Starlink (SAT-01) takes the live programme off the island: it is the mandatory exit to the audience. Iridium (SAT-BK) carries backup data only.

Radio links

Camera	Films	Distance	Estimated signal	Antenna at least
C1-01	CP0 start/finish	636 m	-51.7 dBm	3.7 m
C1-02	CP1	616 m	-51.5 dBm	3.7 m
C1-03	CP2	343 m	-46.4 dBm	3.0 m
C1-04	CP3	709 m	-52.7 dBm	3.9 m
C1-05	CP4	727 m	-52.9 dBm	3.9 m
C1-06	CP5	605 m	-51.3 dBm	3.7 m
C1-07	CP6	264 m	-44.1 dBm	2.8 m
C1-08	CP7	513 m	-49.9 dBm	3.5 m

Calculated at ~2.3 GHz with 250 mW and 3 + 17 dBi antennas (assumption). A signal above about −65 dBm is considered strong. “Antenna at least” = height above shrubs and people to keep the line of sight clear.

Key data

- Track lap: ~3.23 km · technical hub 130 m from the airstrip and 187 m from the track.
- Flat island: maximum ~6 m, shrubs 0.5–1 m.
- Heavy rain: negligible at 2 GHz.
- Capacity: up to 31.7 Mbps per transmitter, enough for 1080p50 HEVC.
- Delay: one frame (manufacturer figure).
- Radio network cost: used HCAM £9,800–11,760 each × 8; Quantum on request.

Legend: yellow = race track · dashed blue = video radio · dotted light blue = towards the satellites (not to scale) · diamond = checkpoint · green = group position · dashed purple = proposal · grey = existing, not verified.